ECON 307 Final Paper **Myunggong Seo**
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Behind the ticket cost: Quantifying the impact of monopoly and operational factor on pricing of US domestic airfare

I. Introduction

Air travel as a service has been the dominant mode of transportation for long-distance travel since the rise of commercial aviation in the United States. Every day, over 3 million passengers travel in and out of US airports. However, following the recession and financial hardships over the last 2 decades, 49 commercial and cargo airlines faced bankruptcies, followed by numerous mergers and industry consolidation, leading to a decreased number of carriers (Transportation Research Board (TRB) 2021). Despite this consolidation, both passenger traffic and capacity have been steadily increasing. In fact, the number of air travelers in 2024 surpassed that of pre-pandemic times, with 67.4 million passengers traveling alone in September in 2025 (Federal Aviation Administration 2024). At the same time, changes in the operational structure led to an increasing number of large aircraft that served fewer routes, and low-cost carriers gradually expanded market share, propelled by a different cost structure (Transportation Research Board (TRB) 2021). Given this context, changes in airfare can account for a significant expenditure for many Americans. Airfare is not only determined by changes in demand, including seasonality, but a multitude of other factors such as the distance of flight, market share of the leading carrier, and cost of fuel. In general, longer travel distances and higher fuel costs drive airfare up. Lower demand can lower them; however, a single carrier occupying more than 80% can distort this pricing dynamics due to its innate market advantage, we believe that the aforementioned factors may affect overall airfare differently than those facing competition. This leads to the research question underpinning this paper: **Do U.S. domestic routes under monopoly have higher overall fares than**

those without? The base group of comparison for our test will be US domestic flights without monopoly and low cost carriers during quarters 1 and 2.

II. Literature Review

When it comes to the difference between monopolistic and competitive routes, Malighetti, Redondi, and Salanti (2014), argue that the number of competing carriers on a route is heavily driven by the overall demand for that specific destination pair. Their empirical findings discovered that average prices were systematically higher on competitive routes compared to those operated on a monopoly. Therefore, a low level of demand for a route is sufficient to force carriers to impose lower fares, regardless of the degree of competition. Work by K. Gülnaz Bülbül (2025) used market level US airfare data with regression and ML models to understand how competition and route characteristics affect airfare. The author found that variables such as distance, number of passengers, presence of low cost carriers, and measures of competition were some of the strongest predictors of average fares. Moving on to the key variables used to estimate the overall fare, a study by Studzieniecki et al. (2025) confirms that "distance remains an important but not dominant" aspect in determining the overall airfare of LCCs. This means that in general, a longer distance is likely to increase the overall fare. Another article from Dong, Atallah, and Galdo (2025) uncovered the impact of fuel cost on airfares. Fuel cost often accounts for a significant part of operating cost. In fact, fuel accounted for 29.13 % of American Airlines'and 44.96 % of JetBlue Airways' total operating costs during the third quarter of 2008. The authors concluded that low-cost carriers (LCCs) were more likely to increase airfare in response to rising fuel costs, and that a 10% increase in fuel prices leads to a 0.172% increase in airfare for legacy carriers and 0.351 % for LCCs (Hortaçsu, Oery, and Williams 2022). Another paper (Clark, Jørgensen, and Mathisen 2011) models transport pricing across various market structures to demonstrate that the impact of competition on airfare is heavily dependent on distance. While competition lowers fares overall, less intense competition actually makes it more likely that fares will decrease as travel distance rises. Furthermore, the research reveals that competitive market structure has significantly less influence on fares on longer routes, which highlights the complex pricing dynamics behind determining fares. Seasonal variation also has a significant effect on airfares, with airlines usually raising prices during peak demand periods (summer and holidays) and lowering them during off-peak periods. The article from Mckinsey (Bouwer, Hausmann, and Lind 2024) underscores the importance of the structural difference in the operating model of airlines during peak and off-peak season, which implies airfares are fundamentally different between the seasons. Airlines possess significant pricing leverage during the summer, given the excess demand and capacity constraints. Lastly, Williams (FTC) shows that dynamic pricing is often used to maximize profit, and that responses to demand and cost changes differ systematically between competitive and monopoly airline routes, which motivates interactions between the presence of LLCs and monopoly routes.

Based on the literature above, this paper will take into consideration some of the key variables mentioned in the literature, such as the number of passengers (demand), distance, cost of fuel, and whether a monopoly exists in the route, along with the presence of low-cost carriers. Also included into the model will be peak and off peak season to reflect the difference in demand each time of year. However, unlike some of the models in literature, the paper will not take dynamic pricing into account as a factor that impacts overall airfare, and will simplify measurement of competition into whether a route is under a monopoly or not.

III. Conceptual Framework and Model

Following the literature review, we construct the following models:

$$fare_{it} = \beta_0 + \beta_1 passengers_{it} + \beta_2 nsmiles_{it} + \beta_3 monopoly_{it} + \beta_4 lccpres_{it} + \varepsilon_{it}$$

$$ln(fare_{it}) = \beta_0 + \beta_1 ln(passengers_{it}) + \beta_2 ln(nsmiles_{it}) + \beta_3 monopoly_{it} + \beta_4 lccpres_{it} + \beta_5 peakseason_{it} + \beta_6 ln(jetfuel_{it}) + \varepsilon_{it}$$

$$ln(fare_{it}) = \beta_0 + \beta_1 ln(passengers_{it}) + \beta_2 ln(nsmiles_{it}) + \beta_3 monopoly + \beta_4 lccpres_{it} + \beta_5 (monopoly_{it} \times lccpres_{it}) + \beta_6 peakseason_{it} + \beta_7 ln(jetfuel_{it}) + \varepsilon_{it}$$

Our models will build on a simple assumption that, holding fuel cost as a control variable throughout different models, then additionally take seasonality into account and interaction between monopoly and the presence of LCCs. The functional forms of major numerical coefficients (nsmiles, passengers, fuel cost) were all logged based on literature review, and confirmation of through visualization in appendix where log form was able to reduce the skewness of relationship. The expected signs for non-stop miles are positive given the overwhelming evidence from literature on how longer flights burn more fuel, leading to an increased airfare given the cost structure for many airlines. Similarly, signs for jet fuel will be positive as the airlines will use a significant part of the revenue from airfare on (Dong, Atallah, and Galdo 2025). Having greater number of passengers implies larger demand for a route, which results in capacity constraint by the airlines as noted by literature, granting them a greater pricing power and therefore a positive relationship with airfare. While the scatterplot of airfare against passengers (Fig. 2 & 3, Appendix) confirms this to some extent, the magnitude sign will not be large. Literature and real world data also shows that summer and holidays are peak demand seasons. Therefore, peak season will likely have a positive relationship with airfare. When it comes to monopoly, literature does suggest despite their upper hand in market dynamics, distance still has a significant effect. However, given we take distance into account, expected coefficient for monopoly is still positive (Malighetti, Redondi, and Salanti 2014). On the other hand, the LCCs can often take lower airfare given their cost structure, and therefore its coefficient will be negatively correlated with airfare (Transportation Research Board (TRB) 2021). The biggest econometric challenge is perhaps endogeneity and reverse causal relationships. Because this is a time-series panel data, often changes in airfare can inversely have an impact on the demand or number of passengers . Furthermore, we do not consider the total number of carriers serving a route, which can have a major impact on measuring level of competition. Instead of the conventional Herfindahl–Hirschman Index used to benchmark market competitiveness (The United States Department of Justice 2024), monopoly on a route is measured using a more simple classification of monopoly of ~80% recognized by the US Department of Justice (2015). This model can also be affected by a multitude of other factors, such as random shocks from geopolitical and natural events. Similarly, this model doesn't take into account the distinct cost structure related to individual carriers and ground service providers, and the number of alternative means of transportation available (Dong, Atallah, and Galdo 2025), which can possibly lead to omitted variable bias.

IV. Data Description

The Consumer Airfare Report is a quarterly generated dataset from the US Department of Transportation, which compiles data upon receiving reports from individual carriers. This specific subset contains the top 1000 most contiguous city pair route pairs from 1996 to Quarter 3 of 2025, with a total of 119,035 observations and 26 rows. All observations are aggregated at route-level, meaning they represent a single route between a city pair each noted by city name and numeric ID. **Fare** stands for overall average fare for that route paid by all fare paying passengers regardless of seat class, in US dollars. However, it does not cover free tickets, such as those awarded by carriers offering vouchers, rewards including frequent flyer programs. **Nsmiles** refers to non-stop miles or the distance between the city pair measured in radians. **Passengers** are the sum of the number of passengers traveled for that route during the quarter. The carrier with the largest market share is denoted **carrier_lg** with two letter

IATA codes with its specific market share **large_ms** in decimal percentages and corresponding average fare **fare_lg** in US dollars. Similarly, **carrier_low** stands for Carrier with the lowest fare, with its market share and fare price **lf_ms** and **fare_low**. **Jetfuel** is the quarterly average price of kerosene based jet fuel source from the US Gulf in Dollars per Gallon without seasonal adjustment (U.S. Energy Information Administration).

Table 1. Summary statistics of all the numerical variables of the data excluding dates

Variable	Obs	Mean	Std. Dev.	Min	Max
fare	119032	199.545	62.746	56.42	676.89
nsmiles	119032	1056.966	609.413	109	2724
passengers	119032	824.71	1315.594	27.582	25471.42
jetfuel	119032	1.737	.896	.335	3.969
large ms	119032	.557	.184	.1	1
fare lg	119032	204.134	72.709	51.49	679.125
lf ms	119032	.323	.238	.01	1
fare low	119032	166.142	58.043	51.49	669.736
monopoly	119032	.121	.326	0	1
lccpres	119032	.62	.485	0	1
lnfare	119032	5.245	.328	4.033	6.518
lnpassenger	119032	6.194	.917	3.317	10.145
lnmiles	119032	6.786	.62	4.691	7.91
lnjetfuel	119032	.387	.617	-1.094	1.379

The summary Table 1 of the numeric variables revealed the magnitude and scale of each variable. No continuous numerical variables had a minimum value of zero. Notably, the number of passengers and nonstop miles had the biggest spread given the large standard deviation of ~1315 and ~609. Jet fuel price (kerosene) had the smallest scale of its given unit of measurement as dollars per gallon. Average market share by the leading carrier for a route was little over 0.55, or roughly about 55%.

Table 2. Correlation matrix of all the numeric variables ex. dates and log forms

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) fare	1.000									
(2) nsmiles	0.537	1.000								
(3) passengers	-0.056	0.015	1.000							
(4) jetfuel	0.249	0.023	0.065	1.000						
(5) large_ms	-0.173	-0.460	-0.161	-0.031	1.000					
(6) fare_lg	0.952	0.511	-0.012	0.212	-0.166	1.000				

(7) lf_ms	-0.337	-0.287	-0.089	0.016	0.316	-0.397	1.000			
(8) fare_low	0.822	0.457	-0.076	0.201	-0.164	0.737	-0.076	1.000		
(9) monopoly	-0.132	-0.276	-0.118	-0.038	0.670	-0.136	0.235	-0.128	1.000	
(10) lccpres	-0.147	-0.058	0.105	0.255	-0.015	-0.148	0.099	-0.219	-0.041	1.000

Correlation matrix revealed that overall fare was highly correlated with fare of carriers with both high market share and lowest price, as expected. Nonstop miles had the next highest positive correlation with fare. Otherwise, no significant positive/ngative correlation was observed between the variables. Interestingly, a decent level of negative correlation existed between market share of leading carrier and the nonstop miles. Upon investigation, only 3 rows of data specific to market share of lowest carrier (lf_ms) and corresponding fare (fare_low) were missing and were dropped given its rather negligible significance to overall number of observations. Also dropped were geocoded city zip codes and addresses from columns 16 through 26 as they were mostly empty. They contain no numerical value nor were they a significant part of the regression. The average quarterly price of kerosene based jet fuel cost per gallon (jetfuel) was appended to the Airfare Report prior to the construction of other variables. 3 dummy variables and 4 logged variables were newly constructed. Monopoly is a dummy variable that takes 1 if a route is determined to be under a monopoly formulated by filtering all routes where the market share for the carrier with the largest market share (large_ms) exceeds 0.8 or 80%. lccpres is another dummy variable constructed to indicate the presence of a low-cost carrier on a route. This was done by using the inlist function of Stata with a given set of LCC IATA codes and looping through both the high and low market share carriers. Lastly, peakseason dummy variables were built to take 1 if the observation took place during Q3 & Q4, which are peak flight demand seasons over late summer and fall & winter holidays. Log of overall fare, passengers, non-stop miles and fuel cost were generated using ln(variable) command.

V. Methodology and Results

Table 3: Regression Analysis Results

Variables	(1) fare	(2) lnfare	(3) lnfare
nsmiles	0.0538*** (0.0002)		
passengers	-0.0029*** (0.0001)		
monopoly	1.582*** (0.460)	-0.0196*** (0.0024)	
lccpres	-23.66*** (0.307)	-0.144*** (0.0016)	
jetfuel	20.14*** (0.165)		
lnmiles		0.259*** (0.0013)	0.263*** (0.0012)
lnpassenger		-0.0304*** (0.0008)	-0.0311*** (0.0008)
peakseason		-0.0253*** (0.0015)	-0.0262*** (0.0015)
lnjetfuel		0.173*** (0.0013)	0.174*** (0.0013)
1.monopoly			0.140*** (0.0036)
1.lccpres			-0.109*** (0.0017)
1.monopoly × 1.lccpres			-0.275*** (0.0046)
Constant	124.6*** (0.446)	3.714*** (0.0104)	3.669*** (0.0103)
Observations	119,032	119,032	119,032
R-squared	0.381	0.382	0.401

Standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01; Our model of interest is (3)

Building on top of our baselines model, the final model that included the interaction term between the monopoly dummy and LCC dummy had the highest adjusted R-squared and smallest MSE. All variables, including the interaction term, were statistically significant ($p < 0.001$ & 0 not in CI). Routes under monopoly had about ~14% higher fares than non-monopoly routes, holding other factors constant. Presence of LCC is correlated with a lower overall fare, regardless of monopoly. However, its magnitude was greater when it was present despite being the monopoly or the carrier with the lower fare (~27%). Every 1% increase in nonstop miles, number of passengers, and jet fuel price is associated with a 0.26% increase, 0.03% decrease, and 0.17% increase in overall fare, respectively holding other factors constant. Routes during peak season were associated with 0.02% lower airfare, holding other factors constant. The null hypothesis that the interaction between those are insignificant $\beta_3 = \beta_4 = \beta_5 = 0$ was rejected ($p < 0.001$)

Based on this, the joint test between the two terms were also significant with Prob > F being less close to 0. Two primary cost variables (nonstop miles & passengers) were also jointly significant ($p < 0.001$). We also reject the null hypothesis that distance is not positively correlated with overall airfare. The R-squared of 0.4 means all the variables were able to explain about ~40% of the variation of overall fare, which is reasonable. The only econometric issue observed was the anomaly of peakseason being negatively correlated with overall fare.

VI. Discussion & Conclusion

This paper aimed to use a standard OLS model to estimate the impact of monopoly and other Overall, the results suggest that airfares rise with distance and fuel costs, marginally fall with passenger volume, and are strongly affected by market structure and low-cost carrier presence. Most importantly, we were able to confirm that monopoly had significant association with increase in overall fare. At the same time, the negative interaction between monopoly and LCC presence suggests that LCCs can act as a ceiling that prevents monopolists from fully exploiting their market position, regardless of the fact whether LCC itself has the largest share for a route. A possible explanation for the marginal negative coefficient of passengers likely reflects economies of scale. Carriers serving high volume routes can utilize larger aircraft or spread fixed costs over more travelers to decrease average fare down despite the high demand (Bouwer, Hausmann, and Lind 2024).

However, our model does come with limitations. Dynamic pricing (Hortaçsu, Oery, and Williams 2022) is one of the key aspects that determines the price of an airfare. Airfares can often differ by magnitude depending on the date and time of purchase prior to departure. However, the Consumer Airfare Report doesn't take such factors into account, which can lead to not only omitted variable bias. Another omitted bias is aircraft type and age being used for that route, which can have significant impact on operating costs and determining capacity constraint (Dong, Atallah, and Galdo, 2025). Most importantly, endogeneity is a central concern in panel-data models because regressors may be correlated with unobserved factors that also affect the dependent variable. Fixed effects can control for unobserved heterogeneity in a time series data, but they do not fully resolve endogeneity when dependent and independent variables are simultaneously determined, which was our model. (Park, Song, and Lee 2020). Additionally, a sustained high fare can either drive the demand away in certain cases, resulting in a reverse causality between airfare and passenger. The negative coefficient behind seasonality could partially explain this issue, as airlines may proactively increase capacity to offset potential upshooting of price (Bouwer, Hausmann, and Lind 2024). Measurement errors could possibly be attributed to nsmiles, considering the occasion where the non-stop distance between the two cities doesn’t perfectly correspond to actual flight distance flown by the aircraft Furthermore, airliners often purchase fuel in bulk ahead to hedge against price volatility, which may add marginal errors on its relation between market price of jet fuel and airfare (Dong, Atallah, and Galdo 2025).

A potential development of this paper can mitigate limitations by introducing additional factors that are closely related with the operational cost and structures of major US airliners, such as aircraft type, size of the airport served, total number of competitors, and the dynamic pricing factor. Because the current models assume a rather simple level of competition, using the HHI with the proper calculation will allow a more realistic classification of routes under Monopoly. A mean fare of ~$199.55 with a 14% monopoly markup translates to roughly an additional $28 per ticket. For a family of four, this could cost over $110, which represents a significant expenditure if accumulated over time. The statistical significance of the result can potentially serve as solid evidence to construct a more robust antitrust framework and rule that can protect the interests of consumers, given the highly consolidated nature of the current US commercial aviation industry.

VII. References

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VIII. Appendix

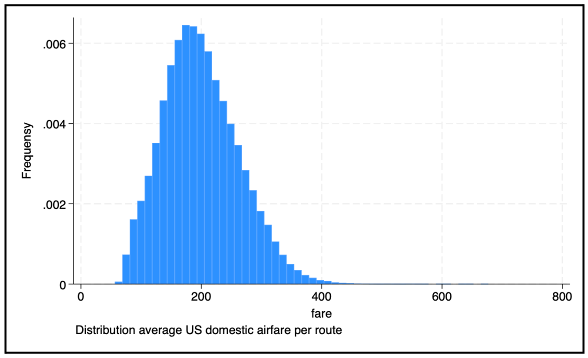

Fig 1.

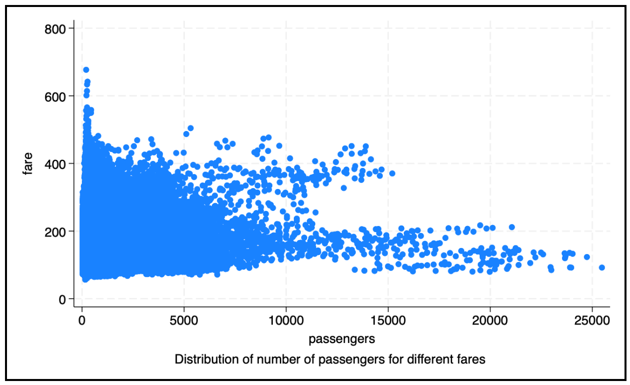

Fig 2.

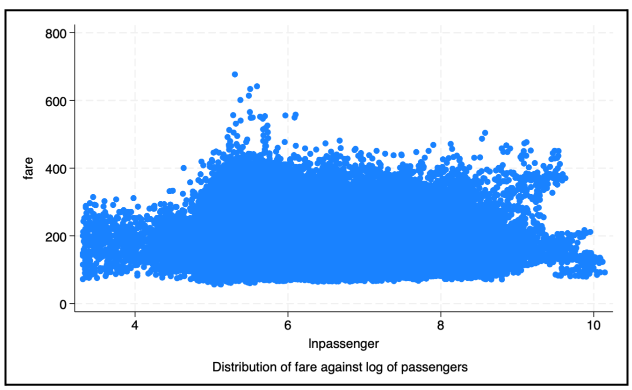

Fig 3.

Fig 4. Historical price of jet fuels

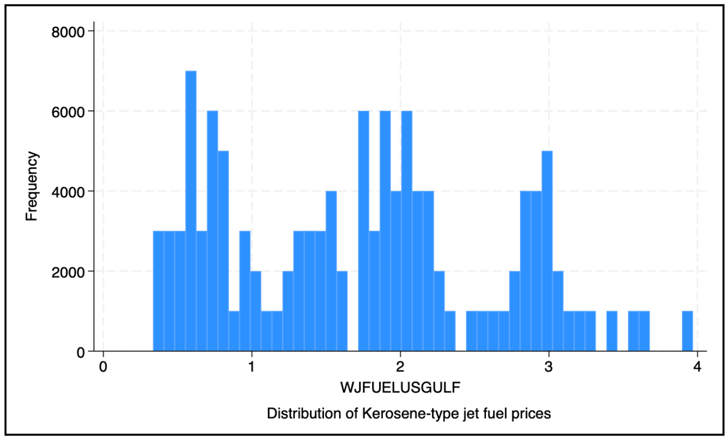

Fig 5.

Source	SS	df	MS			
				Number of obs	=	119,032
				F(5, 119026)	=	14666.92
Model	178656916	5	35731383.1	Prob > F	=	0.0000
Residual	289969810	119,026	2436.18881	R-squared	=	0.3812
				Adj R-squared	=	0.3812
Total	468626725	119,031	3937.0141	Root MSE	=	49.358

fare	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
nsmiles	.0538292	.000245	219.69	0.000	.053349	.0543095
passengers	-.0029459	.0001102	-26.74	0.000	-.0031618	-.00273
monopoly	1.582323	.4597203	3.44	0.001	.6812784	2.483367
lccpres	-23.66151	.3070656	-77.06	0.000	-24.26335	-23.05966
jetfuel	20.1395	.1653338	121.81	0.000	19.81545	20.46355
_cons	124.5742	.4460943	279.26	0.000	123.6999	125.4486

Fig 6. OLS baseline model

Source	SS	df	MS			
				Number of obs	=	119,032
				F(6, 119025)	=	12283.56
Model	4896.18202	6	816.030337	Prob > F	=	0.0000
Residual	7907.15264	119,025	.066432704	R-squared	=	0.3824
				Adj R-squared	=	0.3824
Total	12803.3347	119,031	.107563027	Root MSE	=	.25775

lnfare	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
lnmiles	.2587971	.0012717	203.50	0.000	.2563046	.2612896
lnpassenger	-.0303688	.0008446	-35.96	0.000	-.0320241	-.0287134
monopoly	-.019603	.0024493	-8.00	0.000	-.0244036	-.0148024
lccpres	-.1442904	.0016259	-88.74	0.000	-.1474772	-.1411036
peakseason	-.0253491	.0014956	-16.95	0.000	-.0282804	-.0224177
lnjetfuel	.1730488	.0012738	135.85	0.000	.1705521	.1755455
_cons	3.713895	.0104203	356.41	0.000	3.693471	3.734319

Fig 7. OLS model 2

Source	SS	df	MS			
				Number of obs	=	119,032
				F(7, 119024)	=	11367.95
Model	5130.08744	7	732.869635	Prob > F	=	0.0000
Residual	7673.24722	119,024	.064468067	R-squared	=	0.4007
				Adj R-squared	=	0.4006
Total	12803.3347	119,031	.107563027	Root MSE	=	.25391

lnfare	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
lnmiles	.2628285	.0012545	209.50	0.000	.2603696	.2652874
lnpassenger	-.0311105	.0008321	-37.39	0.000	-.0327413	-.0294796
1.monopoly	.1403644	.0035881	39.12	0.000	.1333317	.1473971
1.lccpres	-.1092317	.0017042	-64.10	0.000	-.1125719	-.1058915
monopoly#lccpres						
1 1	-.2752595	.0045698	-60.23	0.000	-.2842162	-.2663028
peakseason	-.0261617	.0014734	-17.76	0.000	-.0290495	-.0232739
lnjetfuel	.1739141	.0012549	138.58	0.000	.1714545	.1763738
_cons	3.66896	.0102922	356.48	0.000	3.648787	3.689132

Fig 8. OLS model 3

```
( 1)  1.monopoly = 0
( 2)  1.lccpres = 0
( 3)  1.monopoly#1.lccpres = 0

      F(  3,119024) = 3925.83
           Prob > F =    0.0000
```

Fig 9. Joint test result of interaction term in model 3

```
( 1)  lnmiles = 0
( 2)  lnpassenger = 0

      F(  2,119024) =22987.86
           Prob > F =    0.0000
```

Fig 10. Joint test2 result of interaction term in model 3

```
( 1)  1.monopoly + 1.monopoly#1.lccpres = 0
```

lnfare	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
(1)	-.1348951	.0030798	-43.80	0.000	-.1409315	-.1288587

Fig 11. Joint test result of interaction term in model 3

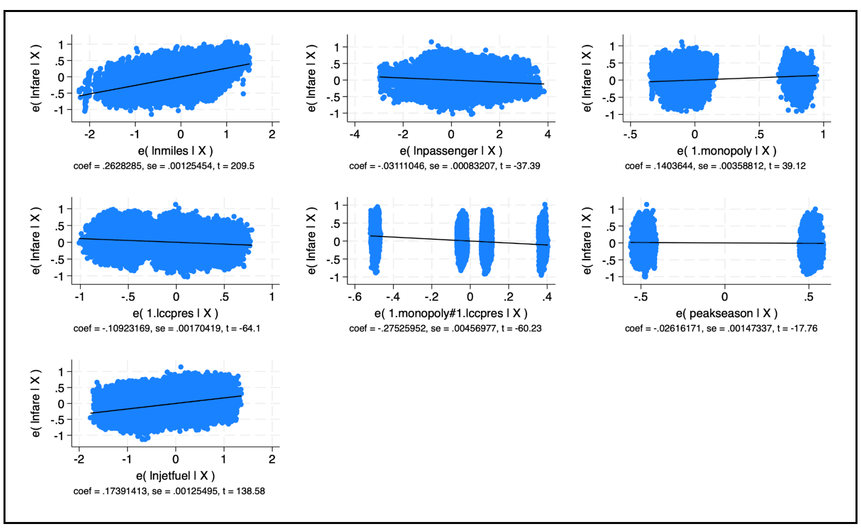

Fig 12. Individual regression out of model 3

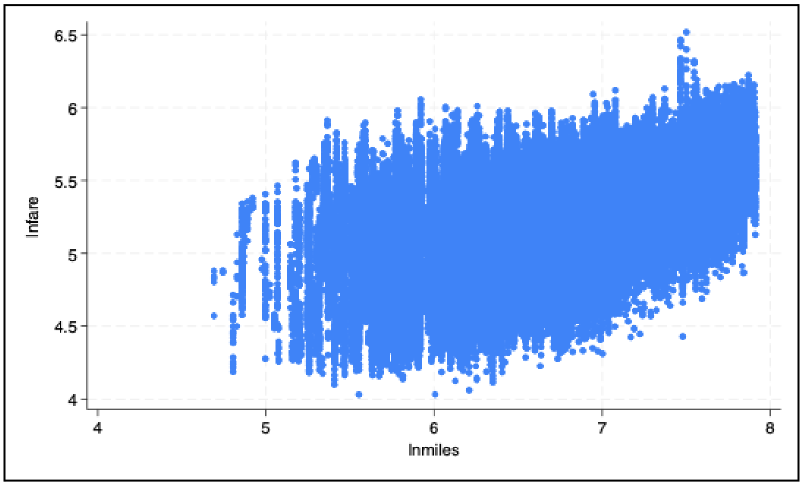

Fig 13. Scatterplot of lnfare vs lnmiles

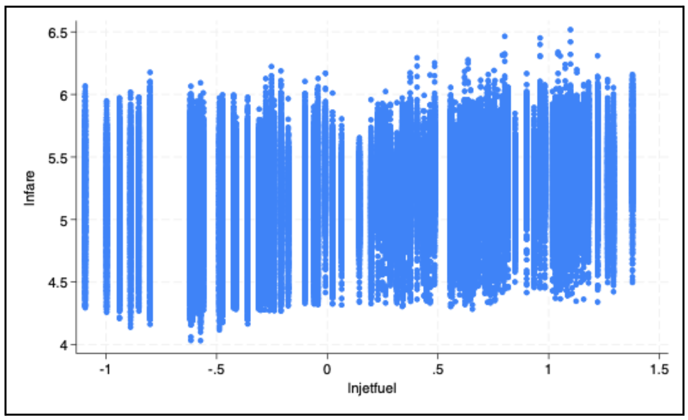

Fig 13. Scatterplot of lnfare vs lnjetfuel

IX. Proposal Feedback Response

The final paper implemented nearly all parts of the feedback. The first change was clarifying the scope and the purpose of the research question, especially about measuring competition. Instead of using a total carrier variable, I chose the presence of a monopoly and the Low Cost Carrier (LCC) to measure the effect of route monopoly on overall air fare. Additional literatures more closely aligned with variable selection and model were chosen and added to the review section. I also fixed a description about the independent variable, especially on b1 would capture the increase in a dollar of average fare for every 1 person increase in passengers per day for a route holding other factors constant. It essentially captures the demand of that route and through logging the variable I allowed the model to capture the scale effect. Furthermore, I expanded the details about how key variables are constructed and other data preparation steps.

Correlation & Multicollinearity was checked through the correlation table to be aware about which variables may be the cause behind the issue. No observations were dropped following the initial data exploration, and all dropped independent variables were placeholder columns with no purpose or influence on the model given their values were absent.

In terms of model, additional functional form, especially log forms were used on fare and several numerical factors based on literature and visual observation since matching a dollar increase to corresponding unit of miles and person can be challenging. Because the number of carriers were not constructed, quantifying competition became more straightforward.

AI Disclosure statement

AI was used during the initial research and idea validation, largely for finding & summarizing the relevant literature that justifies the economic reasoning behind the model. Prompt (Perplexity AI): *I'm currently writing an econometrics paper that models airline ticket price using the department of transportation airfare data. Find peer-reviewed research papers that I can use to justify my variables and functional forms.* Perplexity was also used for cross checking and validation of some of the results from the regressions and connecting them to relevant literature, including limitations.

ChatGPT's codex was used to merge the US WTI Oil prices to the air fare data. Additionally, Google Gemini was used to generate the code replace/inlist Stata command for lcc dummy variable, avgplot command, and extensively refining table outputs for the OLS model.